

ROMA URBE, Italy

WMO: 162350

Lat: 41.952N

Lon: 12.501E

Elev: 79

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 00-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 27.6	(c) 30.1	(d) 15.5	(e) 12.1	(f) 34.8	(g) 18.7	(h) 14.1	(i) 36.7	(j) 19.3	(k) 47.2	(l) 17.3	(m) 47.0	(n) 1.1	(o) 350	(p) 0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 24.4	(c) 94.6	(d) 73.6	(e) 91.6	(f) 72.7	(g) 89.5	(h) 72.1	(i) 77.2	(j) 87.9	(k) 75.5	(l) 86.3	(m) 74.0	(n) 84.6	(o) 8.0	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.8	126.3	82.4	72.0	118.7	81.0	70.3	112.0	79.6	40.7	88.0	38.9	86.1	37.5	84.5	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.6	(b) 14.3	(c) 12.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	45.2	46.7	51.6	57.1	64.4	71.8	76.9	76.7	69.9	63.4	54.4	47.7
	DBStd	12.14	4.90	4.96	5.07	3.93	4.42	5.74	3.78	3.90	5.48	4.43	5.69	5.69
	HDD50	454	162	114	44	2	0	0	0	0	0	0	22	110
	HDD65	2815	612	511	416	238	65	12	0	0	21	83	322	535
	CDD50	4310	15	23	93	214	447	653	833	827	597	414	153	41
	CDD65	1195	0	0	0	0	46	215	368	362	169	32	3	0
	CDH74	12239	0	0	1	21	511	2185	4113	3776	1390	236	6	0
	CDH80	5042	0	0	0	0	106	853	1942	1712	410	19	0	0
(14) Wind	WSAvg	4.3	3.6	4.4	5.0	4.3	4.4	4.7	5.4	5.1	4.4	3.2	3.8	4.0
(15) Precipitation	PrecAvg	33.1	3.5	2.8	2.9	2.5	1.9	1.7	0.6	1.5	3.2	4.4	5.0	4.0
	PrecMax	46.8	6.3	5.1	6.8	8.0	5.1	4.4	2.2	6.7	12.7	13.0	12.1	9.5
	PrecMin	22.4	0.1	0.2	0.6	0.5	0.2	0.0	0.0	0.0	0.0	0.0	0.8	0.7
	PrecStd	6.3	1.7	1.5	1.8	1.9	1.1	1.2	0.7	1.7	2.9	3.8	3.2	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.5	64.5	69.5	76.8	86.4	94.9	97.0	98.3	90.0	82.4	73.7	65.9
		MCWB	55.4	52.5	57.6	62.1	67.8	73.7	75.5	74.4	71.5	68.2	63.5	58.8
	2%	DB	59.3	62.3	66.6	73.2	82.6	91.7	94.7	94.6	87.1	79.1	70.0	62.2
		MCWB	53.3	51.9	55.8	60.4	66.4	72.1	73.5	74.0	69.7	67.1	62.5	57.1
	5%	DB	57.2	59.3	64.7	70.1	79.2	88.9	91.8	91.2	84.4	76.9	67.7	59.4
		MCWB	51.2	50.9	54.8	58.3	65.0	71.2	72.5	73.4	68.3	66.0	60.5	54.5
	10%	DB	55.1	57.1	62.5	67.9	76.7	85.9	89.7	88.2	82.1	74.0	64.7	57.2
		MCWB	49.5	49.6	54.0	57.0	63.7	70.2	72.2	72.0	68.4	64.4	58.5	52.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	56.4	60.4	63.9	70.2	76.4	79.9	78.6	76.4	72.6	66.4	60.2
		MCDB	60.4	61.0	66.5	74.2	80.7	88.0	90.0	89.5	84.6	77.6	70.5	63.9
	2%	WB	55.1	53.8	58.3	61.2	68.4	74.6	77.7	76.9	73.6	69.6	63.8	57.6
		MCDB	58.5	58.6	64.0	70.4	79.0	86.6	88.3	87.8	81.3	75.9	68.2	61.1
	5%	WB	52.4	52.1	56.6	59.7	66.6	72.9	75.6	75.4	71.6	67.4	61.7	55.8
		MCDB	56.3	58.2	62.2	68.1	76.9	85.2	86.7	86.4	80.5	74.3	66.3	59.0
	10%	WB	50.0	50.3	55.1	58.0	64.8	71.1	73.8	73.9	69.8	65.5	59.5	53.3
		MCDB	53.8	56.4	60.6	65.8	74.5	82.7	84.9	84.4	78.5	72.4	64.1	56.5
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	18.9	19.0	20.9	21.5	22.6	24.4	23.7	21.5	20.6	17.7	15.6
		MCDBR	19.8	21.3	23.1	24.5	26.4	26.2	28.1	26.9	25.0	23.9	20.4	17.2
		MCWBR	15.2	13.8	14.6	14.6	12.8	10.4	11.7	11.1	11.5	13.6	14.2	13.0
		MCDBR	17.3	16.9	20.0	22.6	24.4	23.9	24.3	23.7	20.7	20.2	18.2	14.4
	5% WB	MCWBR	15.3	12.4	13.6	14.5	12.7	11.5	12.6	11.8	11.3	13.0	13.6	11.8
(40) Clear-Sky Solar Irradiance	taub		0.349	0.374	0.415	0.444	0.449	0.442	0.433	0.432	0.424	0.411	0.384	0.350
	taud		2.409	2.314	2.195	2.133	2.166	2.241	2.272	2.285	2.302	2.306	2.313	2.403
	Ebn at Noon		249	260	261	262	264	265	266	263	256	243	231	237
	Edh at Noon		26	33	41	47	47	43	42	40	37	33	28	24
(44) All-Sky Solar Radiation	RadAvg		550	839	1228	1634	2001	2280	2342	2042	1490	1015	622	495
(45)	RadStd		55	85	143	119	171	98	108	112	73	88	59	54

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)	+0.72	N/A	N/A	+0.92	N/A	N/A	N/A	-140	+243	+124